

Elia Gatti

Master's Student in Computer Science | HPC & GPU Software Engineer

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Work Experience

Dedagroup - Internship Software Developer

May 2024-Sept 2024

- Reduced VM resource requirements by 50% for the 'TEN' treasury application by **migrating the deployment infrastructure** from a legacy Windows environment to Linux servers.
- Enhanced user experience for internal clients by **developing reactive frontend features** using HTMX and maintaining the Java/Spring backend.

Education

Expected 2026

Master's Degree in Computer Science

University of Trento

Graduated 2024

Bachelor's Degree in Computer Science

University of Trento

Projects & Achievements

Sparse Matrix-Vector Multiplication (SpMV) Optimization [↗](#)

- Developed a **hybrid-adaptive CUDA kernel** for NVIDIA A30.
- Profiled execution time and GFLOPS, focusing on memory access coalescing and occupancy tuning.
- Code available in repository.*

Parallel Minimum Spanning Tree (CPU) [↗](#)

- Implemented parallel MST algorithms (Borůvka / Kruskal) using OpenMP and MPI.
- Analyzed speedup and scalability on multi-core CPU clusters.

Distributed Key-Value Storage System [↗](#)

- Built with **Akka actors** – consistent hashing, data replication, fault tolerance.
- Handled concurrent read/write requests and node failures without client disruption.

MovieMatch – Recommendation Microservices [↗](#)

- Designed modular recommendation backend (Python/FastAPI, Docker Compose).
- Iteratively refactored** from an over-engineered 15-service architecture into a consolidated, maintainable design.
- Implemented API gateway and integrated third-party streaming availability providers.
- (Testing and code quality ongoing – see repository status.)*

DWT-SVD Digital Watermarking Tool [↗](#)

- 1st place** in university competition “Capture the Mark”.
- Implemented embedding using SVD within DWT blocks (Python/OpenCV); built automated attack suite for ROC validation.

Skills

Languages

C, C++, CUDA, Java, Python

Frameworks & Tools

FastAPI, Docker, Git, Linux, OpenMP, MPI

Concepts

GPU programming (memory hierarchy, kernel optimization), parallel algorithms, distributed systems, performance benchmarking